

CS4226

Internet Architecture



Tutorial 1

Question 1

Suppose one is rolling two dice. Which of these pairs of events are independent?

- $E_1 = \text{"First roll is 6"}$ and $E_2 = \text{"Second roll is 6"}$
- $E_1 = \text{"Sum of the rolls is 7"}$ and $E_2 = \text{"Second roll is 4"}$

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- To start with, what is an event?
- What does it mean by saying two events are independent?

Random Experiment

- Consider a random experiment whose outcome cannot be determined in advance
- **Sample space** S : the set of all outcomes
- **Event** E : a subset of the sample space
 - An event E is occurred if the outcome $s \in E$
- **Probability function** $P(E)$
 - $0 \leq P(E) \leq 1$
 - $P(S) = 1$
 - $P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$ for any sequence of events
Given $E_1, E_2, \dots$ that are mutually exclusive

Independence of Events

- Two events are (statistically) independent if the occurrence of one does not affect the probability of the other.

$$P(E_1 \cap E_2) = P(E_1)P(E_2)$$

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- Are the first pair of (E_1, E_2) independent?
 - What about the second pair?

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- Are the first pair of (E_1, E_2) independent?
- What about the second pair?
- What is the sample space?
- What are $P(E_1)$, $P(E_2)$ and $P(E_1 \cap E_2)$?

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- Rolling one dice: $\{1,2,3,4,5,6\}$
 - Each outcome with equal prob. $1/6$.

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- Rolling one dice: $\{1,2,3,4,5,6\}$
 - Each outcome with equal prob. $1/6$.
- Rolling two dice: $\{(1,1), (1,2), \dots, (1,6), (2,1), \dots, (6,6)\}$
 - 36 possible outcomes, each with equal prob. $1/36$.

Question 1

Suppose one is rolling two dice. Which of these pairs of events are independent?

- $E_1 = \text{"First roll is 6"}$ and $E_2 = \text{"Second roll is 6"}$
- First roll is 6
 - The prob. of the subset of events $1/6$
 $\{(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$
- Second roll is 6
 - The prob. of the subset of events $1/6$
 $\{(1,6), (2,6), (3,6), (4,6), (5,6), (6,6)\}$

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- $E_1 \cap E_2$
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 $\{(6,6)\}$
- Therefore, $P(E_1 \cap E_2) = P(E_1)P(E_2)$.

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Suppose one is rolling two dice. Which of these pairs of events are independent?

- E_1 = “Sum of the rolls is 7” and E_2 = “Second roll is 4”
- Sum of the rolls is 7
 - The prob. of the subset of events $1/6$
 $\{(6,1), (5,2), (4,3), (3,4), (2,5), (1,6)\}$
- Second roll is 4
 - The prob. of the subset of events $1/6$
 $\{(1,4), (2,4), (3,4), (4,4), (5,4), (6,4)\}$

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- E_1 = “Sum of the rolls is 7” and E_2 = “Second roll is 4”
- $E_1 \cap E_2$
 - The prob. of the subset of events $1/36$
 $\{(3,4)\}$
- Therefore, $P(E_1 \cap E_2) = P(E_1)P(E_2)$.

Question 1

How about this pair?

- E_1 = “Sum of the rolls is 8” and E_2 = “Second roll is 4”.
- Sum of the rolls is 8:
 - The prob. of the subset of events
 $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$
- Second roll is 4
 - The prob. of the subset of events
 $\{(1,4), (2,4), (3,4), (4,4), (5,4), (6,4)\}$
- Sum of the rolls is 8 and second roll is 4
 - The prob. of the subset of events
 $\{(4,4)\}$
- Therefore, $P(E_1 \cap E_2) \neq P(E_1), P(E_2)$.

Question 2

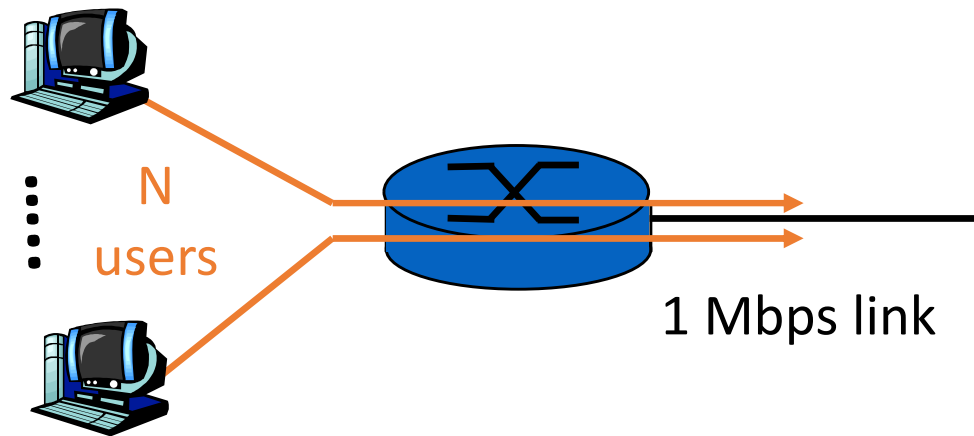
A network link is shared by N users. At any time, each user is online with probability p independently. What is the probability that there are **more than m users being online** at the same time?

Packet switching versus circuit switching

Packet switching allows more users to use network!

Example:

- 1 Mb/s link
- Each user:
 - 100 kb/s when “active”
 - Active 10% of time
- *Circuit-switching:*
 - 10 users
- *Packet switching:*
 - With 35 users, probability > 10 active at same time is less than .0004



Q: How did we get value 0.0004?

Q: What happens if > 35 users ?

Question 2

A network link is shared by N users. At any time, each user is online with probability p independently. What is the probability that there are more than m users being online at the same time?

- Let X be a random variable to indicate the number of users being online. Then $X = n$ means n users are online.
- What is the probability that we want?

Question 2

A network link is shared by N users. At any time, each user is online with probability p independently. What is the probability that there are **more than m users being online** at the same time?

- What is the probability that we want?

$$P(X > m), 0 \leq m \leq N$$

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- What is the probability that we want?

$$P(X > m) = P(X = m + 1) + \cdots + P(X = N)$$

- How to calculate the probability $P(X = n)$?

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- How to calculate the probability $P(X = n)$?

- Independent events E_1, E_2

$$P(E_1 \cap E_2) = P(E_1)P(E_2)$$

- What if we have n independent events?

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- How to calculate the probability $P(X = n)$?

- n users online: $p \cdot p \cdot \cdots \cdot p = p^n$
- $(N - n)$ users offline:

$$(1 - p) \cdot (1 - p) \cdot \cdots \cdot (1 - p) = (1 - p)^{N-n}$$

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- How to calculate the probability $P(X = n)$?

$$P(X = n) = \boxed{\binom{N}{n}} p^n (1 - p)^{N-n}$$

- $P(X > m) = \sum_{n=m+1}^N \binom{N}{n} p^n (1 - p)^{N-n}$

Question 3

The average number of rainy days per year is 10. The weather forecaster is predicting rain for tomorrow. If it rains the next day, 90% of the time the forecaster predicts rain. If it is dry the next day, 10% of the time the forecaster still (incorrectly) predicts rain.

What is the probability that it will rain tomorrow?

Question 3

Conditions:

- The average # of rainy days per year is 10 $\rightarrow Y$.
- The forecaster predicts rain for tomorrow $\rightarrow X$.
- If it rains the next day, 90% of the time the forecaster predicts rain.
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Question:

- What is the probability that it will rain tomorrow?
- (Given the forecaster predicts rain.)

$$P(Y = \text{True} | X = \text{True}) \text{ or } P(Y = 1 | X = 1)$$

Question 3

Conditions:

- The average # of rainy days per year is 10
→ $P(Y = 1) = 10/365$.
- The forecaster predicts rain for tomorrow.
- If it rains the next day, 90% of the time the forecaster predicts rain.
→ $P(X = 1|Y = 1) = 0.9$.
- If it is dry the next day, 10% of the time the forecaster still (incorrectly) predicts rain.
→ $P(X = 1|Y = 0) = 0.1$.

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$$P(Y = 1|X = 1)$$

Conditional Probability

- Conditional probability

$$P(E_i|E_j) \stackrel{\text{def}}{=} \frac{P(E_i \cap E_j)}{P(E_j)}$$

- Joint probability $P(E_i \cap E_j) = P(E_i|E_j)P(E_j)$

- Bayes' Rule

$$P(E_i|E_j) = \frac{P(E_i \cap E_j)}{P(E_j)} = \frac{P(E_j|E_i)P(E_i)}{P(E_j)}$$

Question 3

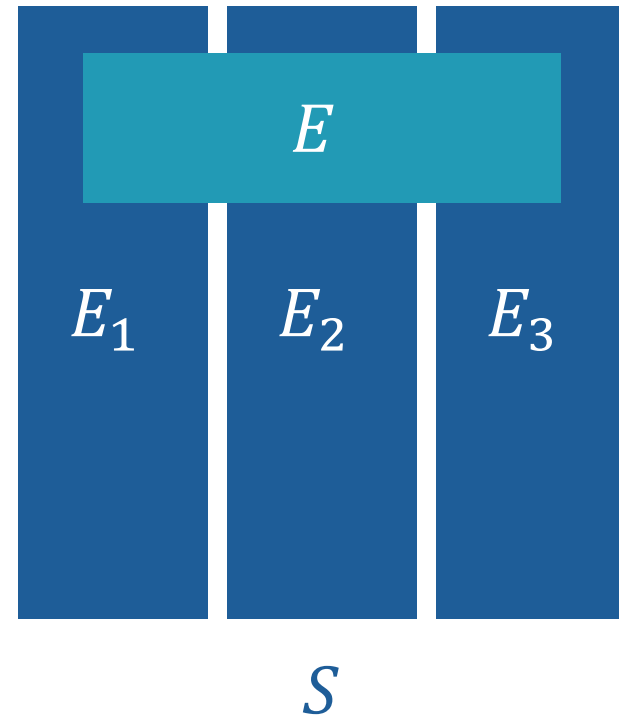
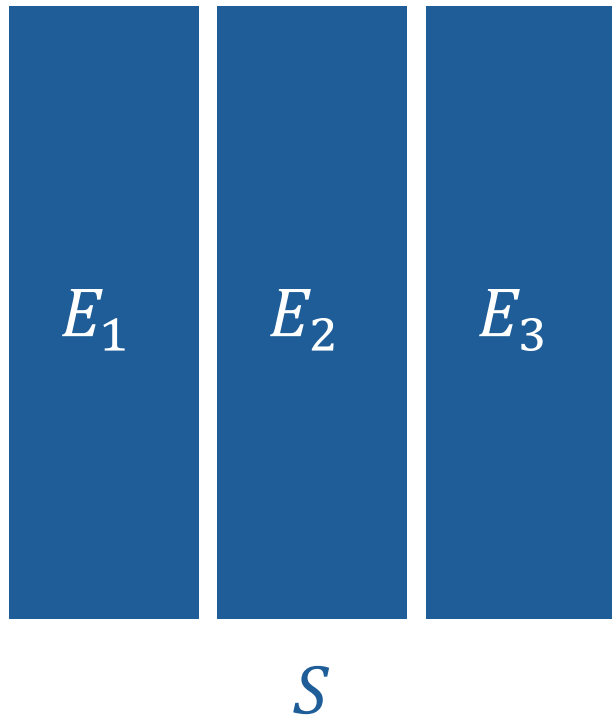
How to calculate $P(Y = 1|X = 1)$? - Bayes' Rule

$$\begin{aligned} P(Y = 1|X = 1) &= \frac{P(X = 1, Y = 1)}{P(X = 1)} \\ &= \frac{P(X = 1|Y = 1) \cdot P(Y = 1)}{P(X = 1)} \end{aligned}$$

How to calculate $P(X = 1)$?

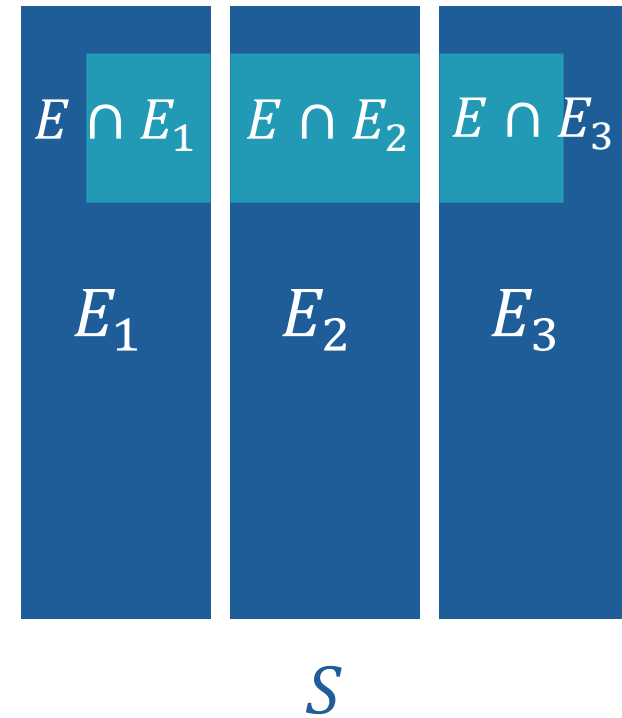
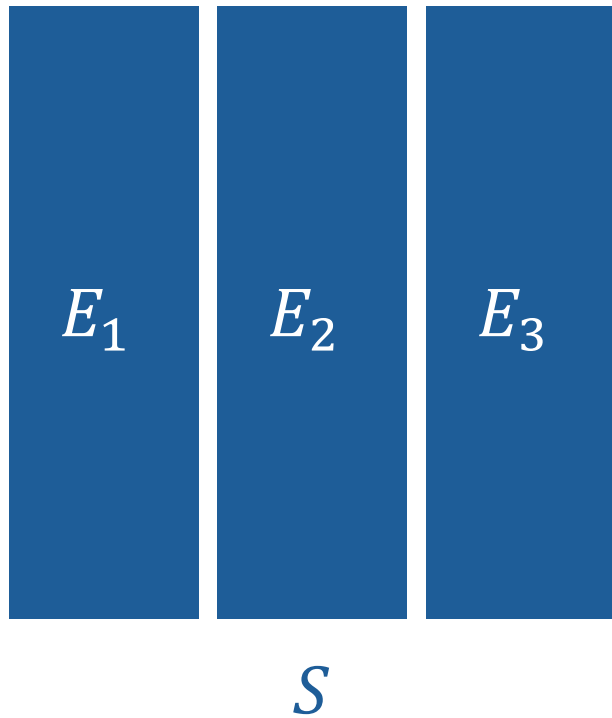
Law of Total Probability

A partition of the sample space:



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Law of Total Probability

- Let events $E_1, E_2 \cdots E_n$ be a **partition** of the sample space, i.e.,

$$\bigcup_{i=1}^n E_i = S \quad \text{and} \quad E_i \cap E_j = \emptyset, \quad \forall 1 \leq i, j \leq n$$

- For any event E , probability $P(E)$ satisfies

$$\begin{aligned} P(E) &= P(E \cap S) = P\left(\bigcup_{i=1}^n (E \cap E_i)\right) = \sum_{i=1}^n P(E \cap E_i) \\ &= \sum_{i=1}^n P(E|E_i)P(E_i) \end{aligned}$$

- A way to “**divide and conquer**”.

Question 3

How to calculate $P(Y = 1|X = 1)$?

$$\begin{aligned} P(Y = 1|X = 1) &= \frac{P(X = 1, Y = 1)}{P(X = 1)} \\ &= \frac{P(X = 1|Y = 1) \cdot P(Y = 1)}{P(X = 1)} \\ &= \frac{P(X = 1|Y = 1) \cdot P(Y = 1)}{P(X = 1|Y = 1) \cdot P(Y = 1) + P(X = 1|Y = 0) \cdot P(Y = 0)} \end{aligned}$$

- $P(Y = 1) = 10/365 \rightarrow P(Y = 0) = 1 - 10/365$
- $P(X = 1|Y = 1) = 0.9$
- $P(X = 1|Y = 0) = 0.1$
- $P(Y = 1|X = 1) = 0.202$

Question 4

A professor uses the following strategy with respect to taking on new students.

- On the even-numbered years, she takes on 2 new students.
- On the odd-numbered years, she takes on 1 new student.

Assuming the average time to graduate is 6 years,
how many students on average will the professor have?

Little's Law : $L = \lambda W$

- Arrival rate into the system λ :

$$\lambda \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{N(t)}{t}$$

- Average sojourn time W :

$$W \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n W_j$$

- Time average # of customers in system L :

$$L \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t L(s) ds$$

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- What are L , λ , W in this question?
 - λ : on average, the professor takes 1.5 students per year.
 - W : the average time to graduate - 6 years.
 - L : number of students on average the professor have. = 9

Thank you.